

CodeWeaver with GPT-5.6 Sol: A Reproducible Evaluation on the ReCodeAgent Benchmark

Abstract

We evaluate CodeWeaver with GPT-5.6 Sol on the 118-project benchmark released with ReCodeAgent (arXiv:26

Evaluation design

Benchmark: 100 CRUST, 6 Oxidizer, 4 AlphaTrans, and 8 SKEL projects across C-to-Rust, Go-to-Rust, Java-t

Artifact completion and claim boundary

Status: COMPLETE
Coverage fraction: 1.0
Unaccounted system cells: 0
System error cells: 0

Evidence inventory

system	rep	expected	measured	accounted missing	unaccounted missing	error
codeweaver	0	118	118	0	0	0
codeweaver	1	118	118	0	0	0
codeweaver	2	118	118	0	0	0
prior	0	118	18	0	0	0
recodeagent	0	118	117	1	0	0

RQ1: Primary CodeWeaver effectiveness (repetition 0)

metric	n	value	95% bootstrap CI	excluded	status
compilation_success	118	100.0%	[100.0%, 100.0%]	0	measured
project_pass_all	118	51.7%	[43.2%, 61.0%]	0	measured
validated_test_micro_pass_rate	118	42.7%	[18.6%, 75.3%]	0	measured
validated_test_macro_pass_rate	118	63.2%	[55.1%, 71.1%]	0	measured

RQ1: Three-repetition variability

metric	n reps	mean	sample SD	95% t CI	status
compilation_success	3	100.0%	0.0%	[100.0%, 100.0%]	measured
project_pass_all	3	51.4%	0.5%	[50.2%, 52.6%]	measured
validated_test_micro_pass_rate	3	42.5%	0.6%	[41.0%, 44.0%]	measured
validated_test_macro_pass_rate	3	63.3%	0.5%	[62.0%, 64.5%]	measured

RQ1: Primary paired comparison with ReCodeAgent

metric	n	CW wins	RCA wins	ties	delta pp	p
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compilation_success	117	37	0	80	31.6	0.0000
project_pass_all	117	27	15	75	10.3	0.0884
validated_test_project_rate	117	12	30	75	-8.7	0.0453

RQ2: Test translation and exact paper tables

The package includes project-level official-comparator evidence, heuristic per-test mappings, translated

RQ3 and RQ4

The Full CodeWeaver protocol is the measured cross-system treatment. No missing CodeWeaver ablation is i

CRUST three-system overlap

Status: measured; triples: 99.

Threats to validity

CodeWeaver and the released baselines are not model-matched fresh reruns; the comparison is observational

Artifact and reproducibility

The result tree contains normalized CSV/JSON/JSONL data, exact paper tables, figures, statistical tests,